MMS Verification Report
`testHighOrderHeatEquationStationary`

Automatically generated from current case results

April 23, 2026

1 Objective

This report documents solver `testHighOrderHeatEquationStationary`, its numerical formulation, and the spatial-convergence study for five MMS examples (`example_0` to `example_4`) in:

- Solver: `/home/pablo/cardiacFoam-pc/applications/solvers/testHighOrderHeatEquationStationary/`
- Case and results: `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/testHighOrderHeatEqu`

2 Governing Equation and MMS

The solver addresses the stationary diffusion problem in $\Omega = [0,1]^3$:

$$\nabla \cdot (\alpha \nabla T) + Q = 0, \tag{1}$$

with constant diffusivity α (here, $\alpha = 1$).

In MMS, an exact temperature T_{ex} is prescribed and the source is set as:

$$Q = -\nabla \cdot (\alpha \nabla T_{\mathrm{ex}}). \tag{2}$$

In this solver, for all five implemented examples, Q is proportional to T_{ex} (Table 1).

Table 1: Exact solutions and MMS source terms implemented in the solver.

Case	T_{ex}	Q
`example_0`	$T_{\mathrm{ex}} = A\sin(\beta\pi x)\sin(\beta\pi y)\sin(\beta\pi z)$	$Q = 3\alpha(\beta\pi)^2 T_{\mathrm{ex}}$
`example_1`	$T_{\mathrm{ex}} = A\cos(\beta\pi x)\sin(\beta\pi y)\sin(\beta\pi z)$	$Q = 3\alpha(\beta\pi)^2 T_{\mathrm{ex}}$
`example_2`	$T_{\mathrm{ex}} = A\cos(\beta\pi x)\cos(\beta\pi y)\sin(\beta\pi z)$	$Q = 3\alpha(\beta\pi)^2 T_{\mathrm{ex}}$
`example_3`	$T_{\mathrm{ex}} = A\cos(\beta\pi x)\cos(\beta\pi y)\cos(\beta\pi z)$	$Q = 3\alpha(\beta\pi)^2 T_{\mathrm{ex}}$
`example_4`	$T_{\mathrm{ex}} = A\sin(kx)\sin(ky)\sin(kz),\ k = \frac{1}{2}\gamma\pi$	$Q = 3\alpha k^2 T_{\mathrm{ex}}$

Parameters used in available results: $A = 1.0$, $\beta = 4.0$ (examples 0–3), $\gamma = 3.0$ (example 4).

3 Numerical Discretization

3.1 Spatial Operator

A high-order diffusive face-flux approximation is used via LRE (Least-Squares Reproducing Kernel) and face quadrature:

$$\phi_f = \int_{A_f} \mathbf{n}_f \cdot (\mathbf{K}\nabla T)\, dA \approx A_f \sum_q w_{f,q}\, \mathbf{n}_f \cdot (\mathbf{K}\nabla T)_q, \tag{3}$$

with $\mathbf{K} = \alpha\mathbf{I}$. Then $\nabla \cdot (\mathbf{K}\nabla T)$ is assembled with `fvc::div(fluxT_HO)`.

When `useHighOrder_T=false`, the solver falls back to `fvc::laplacian(conductivity, T)`.

3.2 Stationary Solution Strategy (Pseudo-Time)

There is no physical time derivative $\partial T/\partial t$. The steady problem is solved by explicit pseudo-time relaxation:

$$T^{k+1} = T^k + \Delta\tau\, R^k, \qquad R^k = \nabla\cdot(\alpha\nabla T^k) + Q. \tag{4}$$

The pseudo-time step is:

$$\Delta\tau = \texttt{pseudoCFL}\,\frac{\Delta x^2}{\alpha}. \tag{5}$$

Iterations stop when $\|R\|_\infty <$ `residualTol` (or `maxPseudoSteps` is reached).

Important note on mass matrices and implicit/explicit time integration. This stationary solver has no mass term (no dT/dt), therefore there is no consistent/lumped mass matrix and no physical-time implicit integration. The method is explicit only in pseudo-time, as an iterative route to the steady state.

3.3 Boundary Stencils

In `createFields.H`, `includePatchInStencils` is enabled only for `fixedValue`/`fixedVoltage` patches. `zeroGradient` patches are excluded from boundary LRE stencils. This can affect near-boundary accuracy and global rates (especially in L_∞-type norms).

4 Error Definitions and Convergence Orders

4.1 Cell-Centered Errors

With $e_i = T_i - T_{\mathrm{ex},i}$ and cell volume V_i:

$$L_1 = \frac{\sum_i |e_i| V_i}{\sum_i V_i}, \tag{6}$$

$$L_2 = \sqrt{\frac{\sum_i e_i^2 V_i}{\sum_i V_i}}, \tag{7}$$

$$L_\infty = \max_i |e_i|. \tag{8}$$

The solver also reports `error_T` as:

$$\texttt{error\_T} = \sqrt{\sum_i e_i^2}, \tag{9}$$

which is not volume-normalized. On uniform 3D meshes, its expected slope approximately follows

$$p_{\texttt{error\_T}} \approx p_{L_2} - \tfrac{3}{2}. \tag{10}$$

4.2 Gauss-Point Reconstructed Errors

At each cell quadrature point $\mathbf{x}_q$, the solver reconstructs:

$$T_h(\mathbf{x}_q) = T_C + \nabla T_C\cdot\mathbf{d} + \frac{1}{2}\mathbf{d}^T\mathbf{H}_C\mathbf{d} + \frac{1}{6}\mathcal{T}_C(\mathbf{d},\mathbf{d},\mathbf{d}), \tag{11}$$

with $\mathbf{d} = \mathbf{x}_q - \mathbf{x}_C$. Hessian and third-derivative terms are included when the LRE order supports them.

Then:

$$G_1 = \frac{\sum_{i,q} V_i w_{iq} |T_h(\mathbf{x}_{iq}) - T_{\text{ex}}(\mathbf{x}_{iq})|}{\sum_{i,q} V_i w_{iq}}, \tag{12}$$

$$G_2 = \sqrt{\frac{\sum_{i,q} V_i w_{iq} (T_h - T_{\text{ex}})^2}{\sum_{i,q} V_i w_{iq}}}, \tag{13}$$

$$G_\infty = \max_{i,q} |T_h(\mathbf{x}_{iq}) - T_{\text{ex}}(\mathbf{x}_{iq})|. \tag{14}$$

4.3 Convergence Rates (`run_convergence.py`)

Using $h = 1/N$, for two consecutive meshes (N_1, N_2):

$$p_{N_1 \to N_2} = \frac{\log(e_1/e_2)}{\log(h_1/h_2)} = \frac{\log(e_1/e_2)}{\log(N_2/N_1)}. \tag{15}$$

Additionally, a global fitted slope p_{fit} is computed by log–log linear regression over all meshes.

5 Run Configuration

- Meshes: $N = 10, 15, 20, 25, 30$ on cube $[0,1]^3$ (uniform N^3 meshes).
- `alpha`=1.0.
- `useHighOrder_T=true`.
- LRE: `N=3`, `Nn=60`, `k=6`, `maxStencilSize=80`, `useQRDecomposition=true`, `useGlobalStencils=true`.
- Pseudo iteration: `pseudoCFL=0.1`, `residualTol=1e-10`, `maxPseudoSteps=200000`.

6 Results: `example_0`

MMS definition. $T_{\text{ex}} = A\sin(\beta\pi x)\sin(\beta\pi y)\sin(\beta\pi z)$ and $Q = 3\alpha(\beta\pi)^2 T_{\text{ex}}$.

BCs in case template. Template in $0/T$: six faces with `fixedVoltage` (homogeneous Dirichlet).
Final convergence summary (mesh $N = 30$):

Table 2: Meshwise errors for `example_0` (file `errors_vs_N_stationary.dat`).

N	L_1	L_2	L_∞	`error_T`	G_1	G_2	G_∞
10	7.048810e-02	1.239924e-01	6.719479e-01	3.920985e+00	9.273971e-02	1.501178e-01	9.332560e-01
15	1.452173e-02	2.395150e-02	1.286408e-01	1.391456e+00	2.120367e-02	3.112282e-02	1.686667e-01
20	5.456393e-03	7.761052e-03	2.780076e-02	6.941696e-01	7.545237e-03	1.027444e-02	3.819653e-02
25	2.262793e-03	3.237674e-03	9.105407e-03	4.047093e-01	3.228864e-03	4.335222e-03	2.612068e-02
30	1.121086e-03	1.576019e-03	5.013332e-03	2.589663e-01	1.606778e-03	2.126752e-03	1.475454e-02

Table 3: Global fitted rates p_{fit} for `example_0` (log–log fit over all meshes).

Metric	L_1	L_2	L_∞	`error_T`	G_1	G_2	G_∞
p_{fit}	3.744	3.967	4.596	2.467	3.683	3.871	3.823

- Pseudo step: $\Delta\tau = 1.111111e-04$
- Pseudo iterations: 1550
- Final residual: $\|R\|_\infty = 9.868018e-11$

Table 4: Pairwise refinement rates for `example_0` ($p_{N_1 \to N_2}$).

N_1	N_2	p_{L_1}	p_{L_2}	p_{L_∞}	p_{G_1}	p_{G_2}	p_{G_∞}
10	15	3.896	4.055	4.077	3.639	3.881	4.219
15	20	3.403	3.917	5.325	3.592	3.852	5.163
20	25	3.944	3.918	5.002	3.804	3.867	1.703
25	30	3.852	3.949	3.273	3.828	3.906	3.133

7 Results: `example_1`

MMS definition. $T_{\text{ex}} = A\cos(\beta\pi x)\sin(\beta\pi y)\sin(\beta\pi z)$ and $Q = 3\alpha(\beta\pi)^2 T_{\text{ex}}$.

BCs in case template. Template in $0/T$: $x_{\min}, x_{\max}$ with `zeroGradient`; faces in y, z with `fixedVoltage`. Final convergence summary (mesh $N = 30$):

Table 5: Meshwise errors for `example_1` (file `errors_vs_N_stationary.dat`).

N	L_1	L_2	L_∞	`error_T`	G_1	G_2	G_∞
10	7.763737e-02	1.274425e-01	6.282979e-01	4.030086e+00	9.923409e-02	1.537929e-01	9.644991e-01
15	1.682324e-02	2.540476e-02	9.723462e-02	1.475883e+00	2.306572e-02	3.285809e-02	3.230757e-01
20	5.759780e-03	8.089638e-03	2.850658e-02	7.235593e-01	7.892973e-03	1.070184e-02	9.012217e-02
25	2.379635e-03	3.354571e-03	1.015775e-02	4.193214e-01	3.341111e-03	4.477018e-03	2.955055e-02
30	1.175889e-03	1.634527e-03	5.346623e-03	2.685803e-01	1.654070e-03	2.191262e-03	1.332941e-02

Table 6: Global fitted rates p_{fit} for `example_1` (log–log fit over all meshes).

Metric	L_1	L_2	L_∞	`error_T`	G_1	G_2	G_∞
p_{fit}	3.811	3.966	4.374	2.466	3.728	3.872	3.973

- Pseudo step: $\Delta\tau = 1.111111e-04$
- Pseudo iterations: 2566
- Final residual: $\|R\|_\infty = 9.974599e-11$

8 Results: `example_2`

MMS definition. $T_{\text{ex}} = A\cos(\beta\pi x)\cos(\beta\pi y)\sin(\beta\pi z)$ and $Q = 3\alpha(\beta\pi)^2 T_{\text{ex}}$.

BCs in case template. Template in $0/T$: faces in x and y with `zeroGradient`; faces in z with `fixedVoltage`. Final convergence summary (mesh $N = 30$):

- Pseudo step: $\Delta\tau = 1.111111e-04$
- Pseudo iterations: 3788
- Final residual: $\|R\|_\infty = 9.953283e-11$

9 Results: `example_3`

MMS definition. $T_{\text{ex}} = A\cos(\beta\pi x)\cos(\beta\pi y)\cos(\beta\pi z)$ and $Q = 3\alpha(\beta\pi)^2 T_{\text{ex}}$.

Table 7: Pairwise refinement rates for `example_1` ($p_{N_1 \to N_2}$).

N_1	N_2	p_{L_1}	p_{L_2}	p_{L_∞}	p_{G_1}	p_{G_2}	p_{G_∞}
10	15	3.772	3.977	4.602	3.599	3.807	2.697
15	20	3.726	3.978	4.265	3.728	3.899	4.438
20	25	3.961	3.945	4.624	3.853	3.905	4.997
25	30	3.866	3.943	3.520	3.856	3.919	4.367

Table 8: Meshwise errors for `example_2` (file `errors_vs_N_stationary.dat`).

N	L_1	L_2	L_∞	`error_T`	G_1	G_2	G_∞
10	9.353306e-02	1.391322e-01	5.201538e-01	4.399746e+00	1.150925e-01	1.692636e-01	1.367801e+00
15	1.847575e-02	2.604867e-02	8.784381e-02	1.513291e+00	2.437095e-02	3.317289e-02	2.807206e-01
20	6.099421e-03	8.617974e-03	3.226959e-02	7.708150e-01	8.222187e-03	1.109065e-02	1.076271e-01
25	2.548152e-03	3.584617e-03	1.474382e-02	4.480771e-01	3.481938e-03	4.660725e-03	4.363761e-02
30	1.258804e-03	1.735708e-03	8.018710e-03	2.852059e-01	1.716143e-03	2.275558e-03	2.345329e-02

Table 9: Global fitted rates p_{fit} for `example_2` (log–log fit over all meshes).

Metric	L_1	L_2	L_∞	`error_T`	G_1	G_2	G_∞
p_{fit}	3.917	3.976	3.775	2.476	3.823	3.911	3.696

Table 10: Pairwise refinement rates for `example_2` ($p_{N_1 \to N_2}$).

N_1	N_2	p_{L_1}	p_{L_2}	p_{L_∞}	p_{G_1}	p_{G_2}	p_{G_∞}
10	15	4.000	4.132	4.386	3.829	4.019	3.906
15	20	3.852	3.845	3.481	3.777	3.808	3.332
20	25	3.911	3.931	3.510	3.851	3.885	4.046
25	30	3.868	3.978	3.341	3.881	3.932	3.406

Table 11: Meshwise errors for `example_3` (file `errors_vs_N_stationary.dat`).

N	L_1	L_2	L_∞	`error_T`	G_1	G_2	G_∞
10	9.537305e-02	1.395818e-01	4.317914e-01	4.413963e+00	1.172532e-01	1.751385e-01	1.262408e+00
15	2.147890e-02	2.952477e-02	1.199727e-01	1.715234e+00	2.669534e-02	3.574145e-02	2.498895e-01
20	6.390003e-03	9.261010e-03	3.894466e-02	8.283299e-01	8.502088e-03	1.141211e-02	6.334695e-02
25	2.703762e-03	3.788625e-03	1.368730e-02	4.735781e-01	3.600480e-03	4.763333e-03	1.831269e-02
30	1.345030e-03	1.826917e-03	8.025505e-03	3.001931e-01	1.774048e-03	2.332400e-03	1.119903e-02

Table 12: Global fitted rates p_{fit} for `example_3` (log–log fit over all meshes).

Metric	L_1	L_2	L_∞	`error_T`	G_1	G_2	G_∞
p_{fit}	3.907	3.954	3.722	2.454	3.828	3.934	4.452

Table 13: Pairwise refinement rates for `example_3` ($p_{N_1 \to N_2}$).

N_1	N_2	p_{L_1}	p_{L_2}	p_{L_∞}	p_{G_1}	p_{G_2}	p_{G_∞}
10	15	3.677	3.831	3.159	3.650	3.920	3.995
15	20	4.214	4.030	3.911	3.977	3.968	4.771
20	25	3.854	4.006	4.686	3.851	3.916	5.562
25	30	3.830	4.000	2.928	3.882	3.916	2.697

BCs in case template. Template in $0/T$: six faces with `zeroGradient`. Final convergence summary (mesh $N = 30$):

- Pseudo step: $\Delta\tau = 1.111111e-04$
- Pseudo iterations: 3276
- Final residual: $\|R\|_\infty = 9.936230e-11$

10 Results: `example_4`

MMS definition. $T_{\text{ex}} = A\sin(kx)\sin(ky)\sin(kz)$, $k = \frac{1}{2}\gamma\pi$ and $Q = 3\alpha k^2 T_{\text{ex}}$.

BCs in case template. Template in $0/T$: $x_{\min}, y_{\min}, z_{\min}$ with `fixedVoltage`; $x_{\max}, y_{\max}, z_{\max}$ with `zeroGradient`. Final convergence summary (mesh $N = 30$):

Table 14: Meshwise errors for `example_4` (file `errors_vs_N_stationary.dat`).

N	L_1	L_2	L_∞	`error_T`	G_1	G_2	G_∞
10	2.396599e-03	3.377504e-03	1.203293e-02	1.068061e-01	3.089920e-03	4.220842e-03	3.560411e-02
15	4.473111e-04	6.193994e-04	2.927116e-03	3.598385e-02	5.975104e-04	7.966491e-04	7.330207e-03
20	1.370265e-04	1.893778e-04	9.206440e-04	1.693846e-02	1.862934e-04	2.472761e-04	2.136154e-03
25	5.442065e-05	7.591736e-05	3.576909e-04	9.489670e-03	7.526875e-05	9.996437e-05	1.029571e-03
30	2.582922e-05	3.604490e-05	1.717897e-04	5.922781e-03	3.594828e-05	4.771781e-05	5.443255e-04

Table 15: Global fitted rates p_{fit} for `example_4` (log–log fit over all meshes).

Metric	L_1	L_2	L_∞	`error_T`	G_1	G_2	G_∞
p_{fit}	4.125	4.132	3.891	2.632	4.054	4.079	3.833

Table 16: Pairwise refinement rates for `example_4` ($p_{N_1\to N_2}$).

N_1	N_2	p_{L_1}	p_{L_2}	p_{L_∞}	p_{G_1}	p_{G_2}	p_{G_∞}
10	15	4.140	4.183	3.486	4.052	4.112	3.898
15	20	4.112	4.119	4.021	4.051	4.067	4.286
20	25	4.138	4.096	4.237	4.061	4.059	3.271
25	30	4.087	4.086	4.023	4.053	4.056	3.496

- Pseudo step: $\Delta\tau = 1.111111e-04$
- Pseudo iterations: 12977
- Final residual: $\|R\|_\infty = 9.956125e-11$

11 Global Comparison Across Examples

Table 17 summarizes representative fitted rates for each example. For a fourth-order spatial operator, expected behavior is $p \approx 4$ in smooth norms (especially L_2 and G_2).

Main observations:

- Across all five examples, L_2 and G_2 slopes are close to 4 (about 3.87 to 4.13), consistent with the high-order spatial objective.
- L_∞ and G_∞ are more sensitive to boundary-local errors and show larger variability.

Table 17: Global summary of fitted rates and pseudo-iteration cost (N=30).

Case	p_{L_2}	p_{G_2}	p_{L_1}	p_{G_1}	p_{L_∞}	p_{G_∞}	pseudo steps
`example_0`	3.967	3.871	3.744	3.683	4.596	3.823	1550
`example_1`	3.966	3.872	3.811	3.728	4.374	3.973	2566
`example_2`	3.976	3.911	3.917	3.823	3.775	3.696	3788
`example_3`	3.954	3.934	3.907	3.828	3.722	4.452	3276
`example_4`	4.132	4.079	4.125	4.054	3.891	3.833	12977

- `example_4` has the smallest absolute errors and rates very close to fourth order.
- `error_T` slopes are around 2.45–2.63, consistent with its non-normalized 3D definition ($p_{L_2} - 1.5$ approximately).

12 Conclusions

The solver correctly implements stationary MMS verification for a high-order LRE-based diffusion operator. Results across the five examples confirm near fourth-order spatial convergence in energy/average norms (L_2, G_2), with max-norm variability attributable to local sensitivity and boundary-stencil treatment (notably, `zeroGradient` patches excluded from boundary LRE stencils).

Referenced Files

- Solver: `/home/pablo/cardiacFoam-pc/applications/solvers/testHighOrderHeatEquationStationary/`
- Field setup: `/home/pablo/cardiacFoam-pc/applications/solvers/testHighOrderHeatEquationStatio`
- Convergence script: `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/testHighOrderHeatE`
- Results: `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/testHighOrderHeatEquationSta`